

AN ILLUSTRATED BENCH GUIDE

Build your Counterframe

Portrait e-ink enclosure for Garden Ink and Open Observatory



4

structural
PLA prints

0

screws, inserts
or glue

1

external lead
into Pi PWR IN

Pi Zero + WaveShare
7.3-inch Spectra 6
HAT (E)

Project photograph: an assembled, working Garden Ink device.

Before you begin

This guide follows the v1.0 engineering-prototype CAD. The photograph does not establish its exact print revision. Physical fit, cable clearance, latch life, temperature and stability still need checking on your build.

Find your place Parts 2 • Printing 3 • Glass 4 • Boards & ribbons 5
Wiring 6 • Closing the case 7 • Power & service 8 • Build record 9

Gather and check the parts

Read this page before printing or ordering a replacement ribbon cable.

Item	What to have ready
Display kit	Waveshare 7.3-inch Spectra 6 HAT (E) : glass panel, dedicated blue driver HAT and separate small FPC adapter. (F), (G) and other driver boards are not covered.
Pi + power	Pi Zero with required GPIO header and microSD installed; suitable Pi 5 V supply and slim, straight micro-USB lead. Check your actual Pi revision and component heights.
Interconnects	Eight-wire SPI/power harness; matched 50 mm overall, 50-way FFC with the kit's correct pitch and contact-side arrangement. Measure the supplied cable; it may need replacing.
Printed parts	One front bezel, carrier, rear cover and stand. Four fit coupons; optional PLA release key. Use ordinary unfilled PLA for the first build.
Bench essentials	Clean protective work surface, calipers, small deburring tool or fine file, and a way to inspect small connectors. A reachable Open Observatory station is needed for live data.

Measure against the CAD assumptions

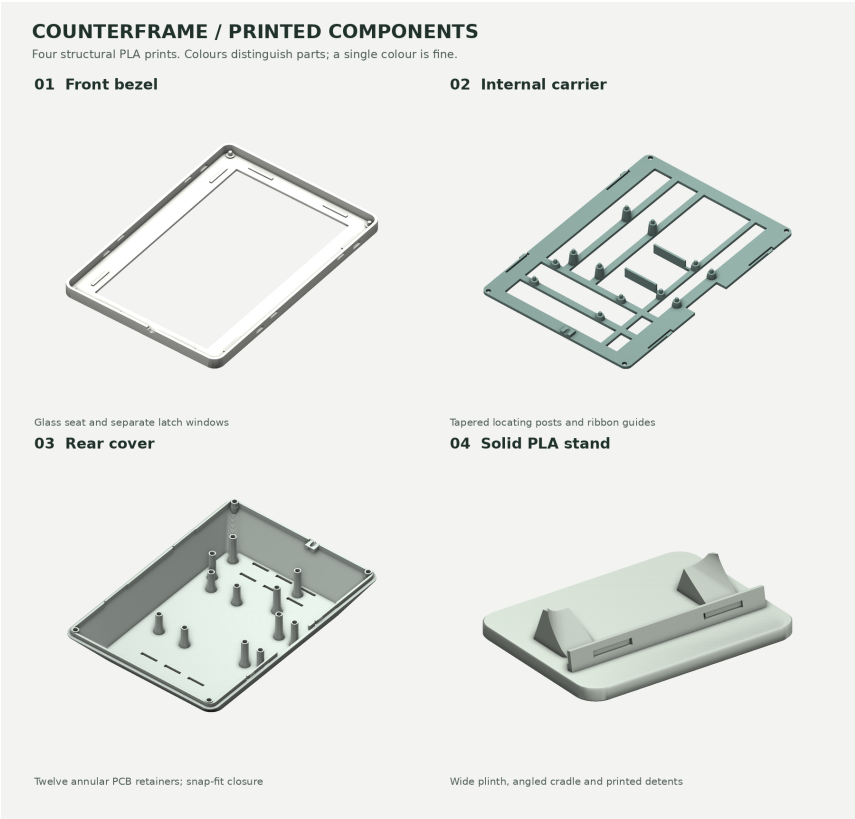
Feature	Nominal model value
Glass / active area	170.2 × 111.2 mm / 160 × 96 mm; model glass thickness 0.91 mm. Verify actual thickness and back features.
Board outlines	Pi: 65 × 30 mm; HAT: 65 × 30.5 mm; adapter: 45 × 21 mm. Check connectors and underside parts.
USB plug / jacket	Moulded plug body 9.6 × 14 × 5.6 mm; cable jacket diameter 3.4 mm. Bulky or right-angle plugs are not an assumed fit.

Check the flexible regions

Identify which parts of the panel tail and FFC may bend. Keep stiffeners, ICs, bonded transitions and seated contact ends straight. Never pull a tail to gain reach.

Prepare the four prints

All supplied manufacturing STLs are in millimetres and already oriented for printing.



CAD parts shown in print orientation. Print one of each from hardware/counterframe/stl/.

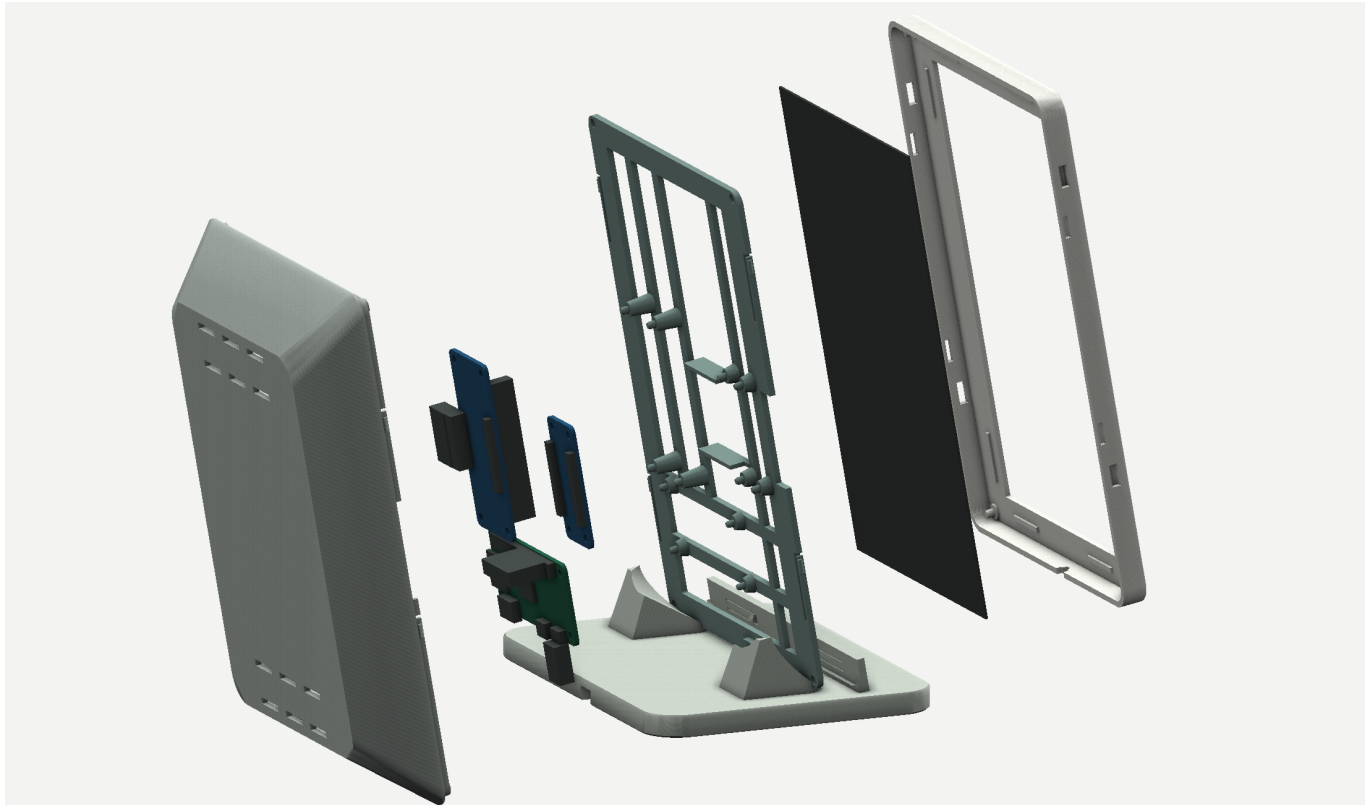
Part / file	Starting settings for a 0.4 mm nozzle
Front · front.stl	Visible face down; 0.16-0.20 mm layers, 4 perimeters, 5 top/bottom layers. No supports intended.
Carrier · carrier.stl	Grid down, posts up; 0.16 mm layers, 4 perimeters, 100% infill . No supports intended.
Rear · rear.stl	Outside face down, collars up; 0.16-0.20 mm layers, 4 perimeters, 5 top/bottom layers. No supports intended.
Stand · stand.stl	Plinth down; 0.16-0.20 mm layers, 4 perimeters, 100% infill . Local supports under front detent leaves/tips.

Test first. Print coupon_pins, coupon_holes, coupon_latch and coupon_socket in the same PLA and fit-critical settings. Check a suitable peg gently against a supported real PCB hole. Reject cracked, whitened or permanently bent latch samples.

Then dry-fit the empty case and stand. Remove stringing, burrs and cable-channel supports. Clips should engage with modest local pressure. Use a 200 × 200 mm usable bed or larger; inspect the slicer preview. Do not scale the parts or print reference electronics meshes.

Seat the glass and carrier

Work face-down on a clean protective surface. Disconnect power before assembly.



CAD exploded view: rear cover → boards → carrier → glass → front bezel. Cables omitted.
The upright view is for orientation; assemble and service face-down.

1 Bench-test before enclosing

Check that the Pi and display operate together. Install the microSD and GPIO header first. Shut down and unplug power before changing connections or fitting the case.

2 Lower the panel into the bezel

Lay the bezel face-down. Place the glass with its image toward the aperture and bonded tail on the **right as viewed from the open rear**. Only the inactive border rests on the seat. Keep dust and burrs away from that border.

3 Engage the carrier's four tabs

Align the carrier with the four bezel locating pins. Keep the tail clear in the relieved right-side pocket. Press the carrier perimeter near each clip, never its crossbars above the glass. Check the panel remains flat and is not squeezed.

The glass is captured, not clamped

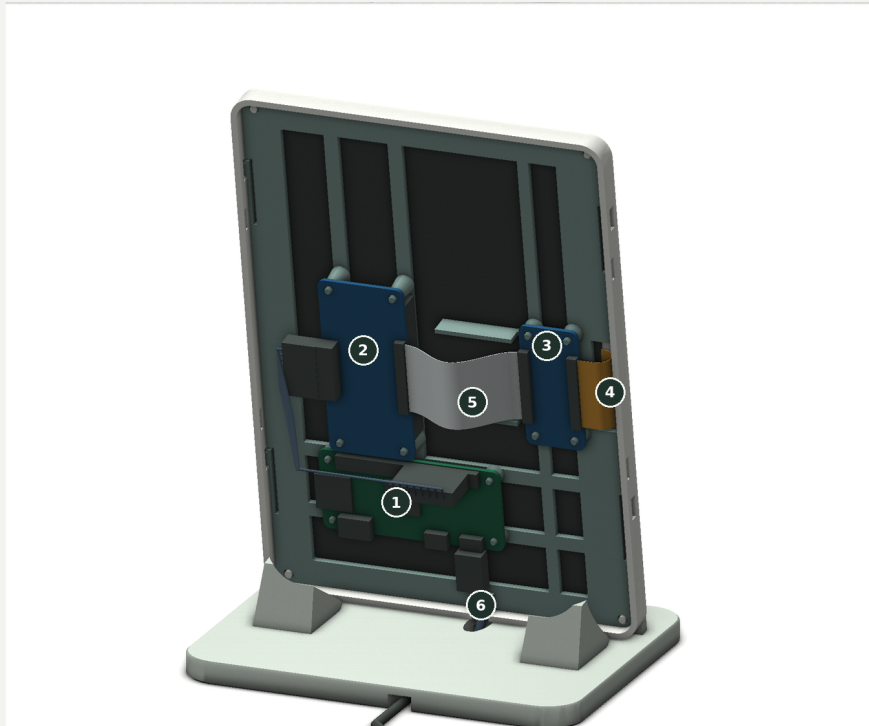
Nominal clearance is 0.30 mm per side and 0.15 mm axially. The aperture centres the image; the unequal glass borders are intentional. If the carrier cannot latch without pressure on the glass, stop and correct the fit.

Fit the boards and ribbons

All left/right directions below are viewed from the open rear.

COUNTERFRAME / INTERNAL LAYOUT

Rear cover removed. Hardware envelopes are approximate; this is not a wiring pinout.



- 1 Raspberry Pi Zero
- 2 Dedicated (E) driver HAT
- 3 Separate FPC adapter
- 4 Bonded panel tail / gentle return
- 5 Matched 50 mm FFC / single arch
- 6 USB jacket saddle / underside exit

One external power lead. No mechanical screws, glue, inserts, metal ballast or elastomer feet.

4 Lower each board onto its four posts

Pi at the bottom; HAT above it on the left; small adapter on the right. Seat every PCB on its standoff shoulders, clear of components and cables. Keep the HAT's factory underside header fitted.

5 Connect the two flexible cables

Return the panel's bonded tail gently into the adapter's outer socket. Connect the matched 50 mm FFC between adapter and HAT in one broad arch. Confirm contact orientation and operate each connector lock as specified for the kit. Keep seated ends straight; no edgewise folds, twists or creases.

Keep the assembly face-down

The boards can lift off their posts until the rear cover is fitted. Modelled bend radii (4.5 mm panel tail; about 6.7 mm FFC) are design assumptions, not approved minimums. Reject a route that bends an unsuitable part of the actual cable.

Wire the HAT and Pi

Power must be disconnected. Use the board’s physical pin numbering, not wire colours.

6 Connect the eight-wire harness

This table is the confirmed project arrangement for the dedicated HAT (E). Identify pin 1 and the header orientation on your actual Pi before connecting. **Physical header pin numbers and BCM GPIO numbers are different.**

HAT label	BCM / function	Physical Pi pin
BUSY	GPIO24	18
RST	GPIO17	11
DC	GPIO25	22
CS	GPIO8 / SPI0 CE0	24
SCLK	GPIO11 / SPI0 SCLK	23
DIN	GPIO10 / SPI0 MOSI	19
GND	Ground	20
VCC	3.3 V in this project setup	17

HAT switch: 0 = 4-line SPI
GPIO8 chip select is controlled by hardware SPI. GPIO18 is not an extra display-power connection. Do not substitute a different Waveshare variant’s wiring.

Route the harness clear of the retainers

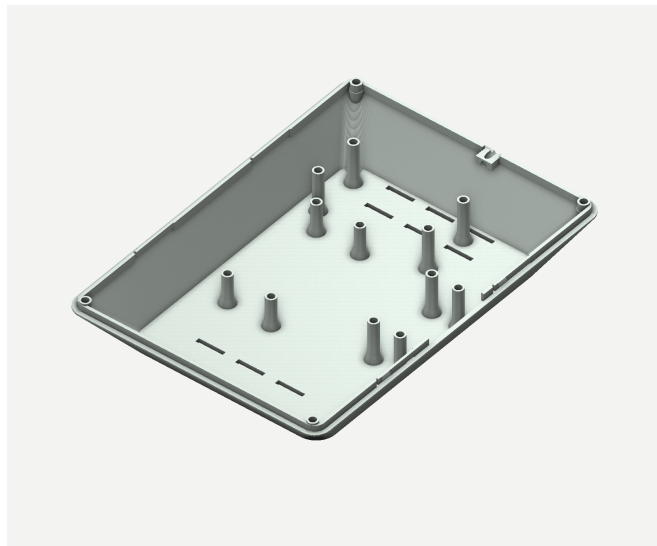
Run wires left from the Pi, **below the HAT’s lower mounting posts**, then up the left side to its eight-pin connector. Keep wires away from all rear-cover collar positions. The coloured paths in the CAD images show routing only.

Connect the one external power lead

Plug the slim micro-USB lead into the Pi’s **PWR IN** socket, not its USB data socket. Place the flexible jacket in the curved lower saddle. External 5 V enters the Pi through PWR IN; the table’s VCC connection is the project’s 3.3 V HAT connection.

Close the case and fit the stand

Check cable clearance before engaging the rear-cover clips.



Left: rear-cover collars, viewed from inside. Right: closed case and underside cable exit. CAD views; board and cable clearances still need checking on your physical build.

7 Lower the rear cover squarely

Keep the frame face-down. The twelve annular collars must surround their pegs and capture the PCB mounting-hole lands, without landing on components, connector boots or wires. Inspect the entire ribbon and USB route, then engage all four cover clips. Do not use the lid to flatten a ribbon or push a connector down.

8 Seat the case in the stand

Place the external USB jacket in the stand's open underside channel. Seat the closed case in the angled cradle and engage both front detents. Check the cable is not trapped and does not lift the base off the worktop. Leave slack behind the stand.

Check the USB saddle

Only the flexible jacket should be lightly captured; the plug and PCB connector must not take a pull. The default groove assumes 0.10 mm squeeze on a 3.4 mm jacket. If compressibility is unknown, start with `cable_grip=0` and regenerate both carrier and rear cover. Never force a mismatched cable into the supplied print.

Power up, inspect and maintain

Garden Ink needs a reachable Open Observatory station for live observations.

- 9 Check the completed assembly**
- Apply power only after checking the wiring and closure. Confirm a full image, cable slack, firm clip engagement and stable placement. Monitor temperature and look for distortion during normal use. The enclosure must not put pressure on the glass.

Stop and inspect if something changes

Disconnect power if the image changes when the frame is gently handled, a connector is under load, a latch cracks or the plastic distorts. Do not push harder to make a tight part fit.

Software setup

For a fresh Pi, follow **docs/operations/DEPLOYMENT.md**: install the display application, check station access, perform one panel refresh and then install the hourly service. The application uses the existing HAT (E) driver. No paid artwork generation is required.

A refresh may wait for the persisted hardware cooldown. Do not clear state or run competing panel demos to force it. When upgrading an existing device, preserve configuration, curated artwork, credentials and all state before replacing files.

Open the enclosure for service

- ☐ Shut down the Pi and unplug power. Support the frame; release both stand detents and lift the case out.
- ☐ Lay the case face-down. Press the four rear-cover tabs inward with the PLA release key, and lift the cover evenly.
- ☐ The PCBs are now free to lift off their posts. Keep the assembly flat and support boards when disconnecting cables.
- ☐ Release the carrier only if the screen itself needs removal. Never pry between the bezel and glass.

Choose a dry, cool location

Keep the vents clear and the frame away from cooking heat, kettle steam, splashes and hot sunlight. The PLA enclosure is not splash-proof or washable. Check stability on your own worktop; PLA can slide on polished or wet surfaces. Lift the finished device by its base.

Check off your finished build

Keep these notes with the printed parts so future fit changes can be traced.

- ☐ Exact HAT (E), Pi revision, board heights, USB plug and FFC contact arrangement checked.
- ☐ Coupons passed; empty parts fitted; no damaged or whitened clips.
- ☐ Glass lies flat with no preload; all four carrier clips are engaged.
- ☐ All three boards sit on their shoulders; connectors are fully seated and locked.
- ☐ Panel tail and FFC remain unstrained; wires clear the cover collars.
- ☐ Four lid clips and both stand detents engaged; cable jacket held without damage.
- ☐ Full image, normal-use temperature, cable slack and worktop stability checked.

Record	Your build
Date / CAD revision	_____
Printer / PLA / layers	_____
Fit changes / cable details	_____
Temperature / fit notes	_____

Sources and further detail

- Construction guide** · [hardware/counterframe/README.md](#)
- Master CAD** · [hardware/counterframe/counterframe.scad](#)
- Project wiring** · [docs/design/MECHANICAL.md](#)
- Pi installation and upgrades** · [docs/operations/DEPLOYMENT.md](#)
- Manufacturer handling and connections** · [Waveshare HAT \(E\) manual](#)

Prepared from the repository's v1.0 CAD and current project notes on 26 September 2026. CAD renders are illustrations of the supplied geometry, not proof of physical fit. This guide adds no hardware qualification. Enclosure CAD/meshes: CC0; photographs, artwork and manufacturer material retain their separate terms.